

00:00–00:30 **Opening** Promise: construct a small error-correcting encoder, then use its Boolean-function counts to understand a quantum gate. One worked example.

00:30–01:30 **1 Memory** Four message bits $\rightarrow$ eight stored bits $\rightarrow$ corrupted readout $\rightarrow$ decoder. At most one error; location unknown. Teaching example. Redundancy adds constraints, not information.

01:30–04:00 **2 Encoder** XOR: equal bits give 0, unequal give 1. Message (b, a_1, a_2, a_3) ; address u . Products mean AND. Hold the message fixed while changing the address. Work 101: $b \oplus a_1 \oplus a_3 = 0$. Reveal 01101001.

04:00–05:30 **3 Affine code** Boolean function = table; affine = constant XOR selected bits. e_i : one bit set. Recover $b = f(000)$ and $a_i = f(e_i) \oplus f(000)$. Therefore 16 distinct codewords in C , among 256 readouts.

05:30–07:30 **4 ANF** First show parity p and readout r . Error at 000: $(1 \oplus u_1)(1 \oplus u_2)(1 \oplus u_3)$. Expand to describe r . Degree = largest product size, not error count. ANF describes; decoder need not expand it.

07:30–09:30 **5 Derivative** Point to 000, 100: comparison changes $1 \rightarrow 0$. Then $D_{e_i}f(u) = f(u \oplus e_i) \oplus f(u)$. Input displacement is not a memory error. Only one of four disjoint comparisons is spoiled.

09:30–11:30 **6 Decode** Affine derivative equals a_i . Votes 0, 0, 1, 1, 1. **Ask why majority works; wait 7 seconds.** Recover all $a_i = 1$; subtract their XOR contribution. Residual 10000000: seven votes give $b = 0$.

11:30–13:15 **7 Intersections** Support = ones; weight counts them. Two faces: $4 + 4$, common edge 2. XOR removes overlap twice: $d_H(f, g) = \text{wt}(f \oplus g)$. Example $d_H(u_1, u_2) = 4 + 4 - 2(2) = 4$.

13:15–15:00 **8 Half count** Choose an occurring input bit. Four disjoint pairs, one one per pair. Name affine hyperplane afterward. Parity-zero contains zero; parity-one is a translate. *Coordinate-change question $\rightarrow$ backup.*

15:00–16:45 **9 Minimum distance** Distinct affine differences have weight 4 or 8; a pair attains 4. Hence $d_{\min}(C) = 4$. Two candidates within one error of r would imply $4 \leq d_H(f, g) \leq 2$.

16:45–18:00 **10 Recap** Encode; describe corruption; recover coefficients; prove uniqueness. $d_H(p, r) = 1$ is one corruption; $d_{\min}(C) = 4$ is code separation.

18:00 checkpoint: classical story complete. Sixteen words form eight complementary pairs. Each pair will represent one quantum basis state.

18:00–19:30 **11 States and phase** Define $|0\rangle, |1\rangle$ before sums. Amplitudes have squared magnitudes summing to one. Relative sign changes the state; common overall phase does not. Bit flip exchanges basis vectors.

19:30–21:00 **12 Physical rotations** Circle = unit complex multiplier, not Bloch sphere. T : $+\pi/4$ on the $|1\rangle$ amplitude; $T^\dagger$: inverse. $|v\rangle$ labels physical bits. In 00101, two ones give two factors. Angles add.

21:00–23:30 **13 Encoding** Start with $a = 000$: normalized sum of all-zero and all-one words. Generally $|a\rangle_L = (|f_{a,0}\rangle + |f_{a,1}\rangle)/\sqrt{2}$. Eight disjoint pairs give eight orthogonal states: three logical qubits, eight physical. **Both words need the same phase.**

23:30–25:00 **14 Desired gate** Only logical label 111 changes sign; now name CCZ. Parity selects the physical $T/T^\dagger$ pattern. For word p , four ones at inverse sites give -4 . Define $W(v) = \text{ones at } T \text{ minus ones at } T^\dagger$; phase $e^{i\pi W(v)/4}$.

25:00–27:30 **15 Check all pairs** Per-site identity gives $W(v) = d_H(p, v) - 4$. Other labels: counts 0, 0; label 111: $-4, +4$. **Ask whether $+4, -4$ give different phases; wait 7 seconds.** Both give -1 . Equal phase within each pair; label-dependent phase implements CCZ.

27:30 checkpoint: toy gate complete. The gate works. The next construction adds correction of an arbitrary one-qubit error.

27:30–28:15 **16 Choose sites** Announce six-bit addresses: u now has four bits; w has two. Four blocks of 16; delete $w = 00$. $64 - 16 = 48$ physical qubits. Geometry alone does not specify the code. *Why toy insufficient? $\rightarrow$ backup.*

28:15–29:45 **17 Weights** Nonzero linear function has 32 ones. If any u bit occurs, deleted block has 8 ones: 24. Otherwise only w bits occur; deleted block has 0: 32. Examples u_1, w_1 . Ordinary divisibility does not prove the signed test.

29:45–30:45 **18 Stabilizers** Six coordinate words S_i ; 64 XOR combinations S . Independence follows because no nonzero linear combination has weight zero. Equal sum has amplitude $1/8$. Flipping mask S_i permutes summands: unchanged state. Name stabilizer after showing this.

30:45–31:45 **19 Logical words** Three chosen words K_i ; $K(a) = a_1K_1 \oplus a_2K_2 \oplus a_3K_3$. Show shift by K_1 first. Eight disjoint collections is a verified choice, not automatic. $|a\rangle_L = \frac{1}{8} \sum_{v \in S} |K(a) \oplus v\rangle$. Each S_i still permutes the terms. *Actual functions? $\rightarrow$ backup.*

31:45–32:30 **20 Rotation pattern** Blocks have 6, 6, 10 inverse sites: $22T^\dagger + 26T$. Read row/column Gray order. ***W now uses this fixed 48-site pattern. Pattern formula? $\rightarrow$ backup.***

32:30–33:30 **21 Integer expansion** Generator row = one of the six S_i or three K_i . Products select common ones. Pair identity first, then triple: coefficients 1, $-2, 4$. At three common ones: $3 - 6 + 4 = 1$. XOR and ordinary addition are different.

33:30–35:30 **22 Exact phase test** Phase depends on W modulo 8. Distinct rows g_1, g_2, g_3 :

$$\begin{aligned} W(g_1) &\equiv 0 \pmod{8}, \\ W(g_1g_2) &\equiv 0 \pmod{4}, \\ W(g_1g_2g_3) &\equiv \begin{cases} 1 \pmod{2} & \text{logical triple,} \\ 0 \pmod{2} & \text{otherwise.} \end{cases} \end{aligned}$$

Higher terms have coefficients divisible by 8. Hence $W(K(a) \oplus v) \equiv 4a_1a_2a_3 \pmod{8}$ for every $v \in S$. Same phase on all 64 words; logical CCZ.

35:30–37:00 **23 Protected result** 48 physical, 3 logical. Define d_Q : fewest sites in an undetected nontrivial logical bit/phase error. Paper proves $d_Q = 3$ using **all** checks; arbitrary one-site correction. Gate test does not prove distance. *Is 48 minimal? $\rightarrow$ qualified backup.*

37:00 checkpoint: own result complete. Return to the classical example for the broader applications.

37:00–39:00 **24 Reed–Muller** Name $\text{RM}(1, 3)$; generalize to degree-bounded evaluations. Linear = closed under XOR. $[n, k, d_{\min}]$: stored bits, message bits, distance. Each codeword has 9 zero/one-error readouts; disjoint sets imply $16 \times 9 = 144 \leq 256$.

39:00–40:45 **25 Erasures/capacity** Missing value, known position. Rate = information per transmitted bit. Capacity = reliable-rate limit as blocks grow for a specified random channel. Suitable growing RM sequences achieve the independent-erasure limit under **optimal whole-block decoding**. No practical-decoder claim.

40:45–43:00 **26 Analysis/learning** Seven of eight agreements, equally likely inputs. Fourier analysis measures parity correlations; recall derivatives for input changes. Learning: examples $\rightarrow$ predictions on new inputs; suitable structure helps. Cryptography resists unwanted prediction. Cubic degree alone did not prevent parity approximation. No transform derivation.

43:00–45:00 **27 Contributions** Repeat: explicit protected encoding; $26T + 22T^\dagger$ physical implementation; exact signed-intersection gate test. Opening counts prepared the final calculation. **Leave contributions visible.** Sources backup only if asked.

If late: shorten applications narration. Preserve decoder assumptions, both phase checks, and the distance/gate distinction. Backups are outside the 45-minute budget; every return button restores the anchor's final reveal.